D035 RESTRAINED APPARATUS

Sommes de Gauss cubiques et revêtements de SL(2), d'après S. J. Patterson

Authority. The 34-page Bourbaki no. 539 scan (SHA-256 `B65B3980...C0F`) controls. All physical pages are article pages and map one-to-one to printed 244–277. P. Deligne is the sole printed author; Patterson is subject attribution. The comparator and every inherited branch remain comparison-only or ZERO_ACCEPTED.

Authority physical page 1 / printed 244. The opening scan reads "SOMMES DE GAUSS CUBIQUES ET REVÊTEMENTS DE SL(2)," with the comma after SL(2), followed on the next line by "D'APRÈS S.J. PATTERSON." The printed byline is "par P. DELIGNE"; Patterson is the subject attribution, not a joint author. The upper-right folio is 539-01 and the date is "juin 1979 (*)". The revision note "(*) texte remanié en juillet 1979." remains at the foot of this page. The displayed product in the Hilbert-symbol formula is indexed by $v|a$. The complete uncropped authority bitmap and a separate presentation derivative accompany the page record and preserve underlining, lineation, mathematical placement, and folio.

Authority physical page 2 / printed 245. The source header visibly reads “359-02”; it is retained rather than silently normalized to the surrounding 539 sequence. The additive character e is underlined in the scan. The authority gives F = $Q(\surd - 3)$, “1mod 3” and “1mod3” without inserted spaces in the indicated lines, and the two displayed *conjugation/reciprocity* relations involving $(\overline{\pi}/\pi)$. The inherited TeX regularizes several of these details; none of those regularizations was accepted without reinspection of the authority. The page ends mid-sentence after $g(\mathfrak{p})^n$ and continues on physical page 3.

Authority physical page 3 / printed 246. The opening continues the sentence from physical page 2. The cyclotomic-field notation is reproduced as printed, $Q(\sqrt[n]{1})$. Equations (0.5)-(0.8), including the absolute-value bars in (0.5), the factor $N(\mathfrak{a})^{-s}$, and the central-extension law $(a, \zeta')(b, \zeta'') = (ab, (a, b)\zeta'\zeta'')$, were replayed from the scan. The authority uses the source spelling “coeur” and the phrase “si impose” later on the page ; the English reader supplies a faithful grammatical translation without altering the French record. The final words “[ ] scinde” continue onto physical page 4. Equation (0.5) intentionally preserves a mixed-case source reading : the summation condition is $N\mathfrak{a} \leq$ x (lowercase x), whereas the right-hand error term is o(X) (uppercase X). The intermittent P01-P05 pixel audit restored this distinction after the earlier replay had silently capitalized the bound.

Authority physical page 4 / printed 247. This page completes the splitting statement from physical page 3 and defines local and restricted-product ε-functions. Formula (0.9), the summation over $(O_\nu/(a))^*$, the series $\sum_{\lambda\in F^*} f([\lambda])N\lambda^{-s}$, and the two Iwasawa matrices were kept in their source order and alignment. The scan identifies $\mathrm{PGL}(2, C)/U(2)$, uses (a z, $|a|v$) for the diagonal action, and states that $\Gamma(3)$ consists of matrices congruent to 1 modulo 3. The inherited candidate changes parts of this model and is recorded only as a post-freeze comparator.

Authority physical page 5 / printed 248. The opening matrix γ and Kubota character $\chi(\gamma) = (c/a)_3$ are retained as displayed. In the local data for τ, the source prints an ε^2-function, the relation $\tau_\nu(x[d^3]) = \tau(x)\|d\|_\nu$, and the expression $\|a\|_\nu f_\nu(a)^-$; the superscript minus is preserved diplomatically rather than expanded or reinterpreted. All four congruence classes in clause (e) are present. Equation (0.11) has constant term $\sigma\nu^{2/3}$, $\sigma = 2^{-1}3^{-3/2}$, and a sum over $\mu \in F^*$. The barred $\chi(\gamma)$ in the Eisenstein summation is also retained. The inherited TeX diverges materially in several of these formulas and was not used as authority.

Authority physical page 6 / printed 249. The opening phrase reads “A des facteurs élémentaires, Γ et ζ près” ; those symbols are retained. The source uses “au pis un pôle simple.” The page defines $E_{s,\Gamma}$, the representation space $\mathscr{E}$, the Fourier coefficient $W^*(f)$, the Fourier expansion, and the covariance relation for W. Matrix placement and the underlined K and e are preserved by the accompanying full-page authority bitmap. Physical page 6 ends with the covariance formula and printed folio 249 ; the Introduction continues on physical page 7, which is outside P01 and has not been processed.

Authority physical page 7 / printed 250. The page continues the covariance statement for W from physical page 6 with “pour $\nu \in$ F.” It then states the T-family result for multiplicative ε^2-functions $\tau' = \prod \tau'_\nu$ and displays $\theta(\tau'; w)$, including the sum over $\mu \in F^*$ and the diagonal matrix $\mathrm{diag}(\mu, 1)$. The authority explicitly says that the method depends on n being odd and on $(n-1)/2 = 1$. The inherited TeX omits the T-family paragraph and replaces that final condition by a different assertion ; neither change was accepted. The page closes the Introduction and ends at printed folio 250.

Authority physical page 8 / printed 251. The heading is “O.O Notations :”. The page fixes Z, T, $B^{\pm}$, and $U^{\pm}$ by four displayed 2 × 2 matrix patterns, then defines the *local*/*global*, adelic, covering, ε-representation, and admissibility notation used later. The source visibly contains the repetition “l’anneau de ses de ses entiers” ; it is retained rather than silently corrected. The “Variante” line reads F_v, O_v. Exact underlining and matrix alignment remain in the full authority bitmap.

Authority physical page 9 / printed 252. This page opens “1. Extensions centrales de SL(2)” and contains Definition 1.1, the Kummer-cohomology construction of Example 1.2, the canonical extension $G(F)^\wedge$, and Remark 1.4. The lower exact sequence includes the symplectic analogue ${}_{-1}L_2(F)$, the kernel ${}_{-1}L_2'$, W, and the map r to $K_2(F)/2K_2(F)$. The inherited TeX substantially replaces the authority’s 1.2-1.4 material and does not preserve this page-local sequence ; the authority wording and displayed arrows control.

Authority physical page 10 / printed 253. Sections 1.5-1.8 define $\mathrm{GL}(2,F)^s$, the commutator and canonical unipotent lifts, and d_1, d_2. The printed text says that the lift in 1.7 is "unique par 1.3" and that the commutator formula determines the extension considered "de $\mathrm{SL}(2,F)$"; both readings are retained without emendation. The page ends with the two matrix definitions of $e^+(a)$ and $e^-(a)$ under "1.9 Formulaire". In the standalone English, the French technical term "sous-groupe distingué" is rendered as "normal subgroup"; the earlier literal "distinguished subgroup" was a false friend. The diplomatic French record remains unchanged.

Authority physical page 11 / printed 254. The page continues the 1.9 formula sheet, including w(a), $d_2(a)$, h(a), identities (1.9.1)-(1.9.5), Example 1.10, and the beginning of 1.11. In the displayed matrix lifted by w(a), the authority shows a^{-1} in the lower-left entry without the minus sign inserted by the inherited TeX; the authority reading is retained. The page ends exactly after (1.11.2), $H^0(A^2 - 0, K_2) = K_2(F)$.

Authority physical page 12 / printed 255. The page gives (1.11.3), the residue formula, the canonical torsor construction, the Čech cocycle {X,Y}, the normalizations of d_1 and d_2 at (0,1) and (1,0), and the opening of 1.13 on the dual affine plane. The source spells “s’annulle” and later “correspond au dual”; both are preserved. The inherited TeX rewrites and abbreviates the torsor argument, whereas the authority includes the residue comparison aX^{-1} versus X^{-1}. The page ends mid-sentence at the locus $\ell(x) = 1$.

Authority physical page 13 / printed 256. The page completes 1.13 from physical page 12 and gives the full 1.14 “Formulaire,” including the definitions of P, $P^\vee$, the four local sections, the two coordinate symmetries, and formulas (1.14.1)-(1.14.3). The lower portion of the authority page is intentionally blank. The inherited TeX compresses the definitions to the three numbered formulas and omits the printed setup and page topology ; those omissions were not accepted. The comparator confirms the same page content and blank remainder.

Authority physical page 14 / printed 257. Section 2 begins here. The authority states $a, b_n = a, b_{nm}^m$ and, in (2.4.1), retains the exponent $(q-1)/n$ after the reduction expression. The inherited TeX gives a different compatibility relation and drops that exponent from (2.4.1) ; both divergences were rejected. The page ends mid-sentence in Rappel 2.7(ii), after “le dual de Pontrjagin.”

Authority physical page 15 / printed 258. The opening line completes Rappel 2.7(ii). The authority then visibly repeats the label “(ii)” and the isotropic-subgroup statement, including the printed quotient A^*/A^* in that repeated line; this apparent source *duplication*/*typographical* anomaly is preserved rather than silently repaired. Example 2.8 prints $Q(\surd-3)$, not the candidate’s $Q(\sqrt[3]{1})$. The large blank remainder and folio 258 are part of the page. The inherited TeX suppresses the repeated passage and normalizes the field notation; neither intervention was accepted.

Authority physical page 16 / printed 259. Section 3 begins. In 3.2 the authority displays the two unipotent matrices ((1 a),(0 1)) and ((1 0),(a 1)); the inherited TeX substitutes different matrices. Diagram (3.3.1) is preserved as an exact-sequence diagram with the diagonal splitting arrow from $\mathrm{SL}(2, F)$. In 3.4 the authority uses the complex archimedean factor $\mathrm{SL}(2, C)$, $\mathrm{PGL}(2, C)$, whereas the candidate replaces it by a real factor. Authority pixels control all of these points.

Authority physical page 17 / printed 260. The page completes Scholium 3.5, then gives 3.6-3.10 : the two constructions of $\mathrm{GL}(2, A)^{\sim}$, the torsor P_n and its distinguished trivializations, the scalar action, the dual torsor, and the notation $P, P^{\vee}$. The authority locus is $\ell(x) = 1$. The inherited TeX replaces 3.5-3.7 by a placeholder summary and changes ℓ to δ ; this is a substantive omission and notation change, so the authority text was retained in full.

Authority physical page 18 / printed 261. The page consists solely of 3.11, which adelizes the local torsor construction and records the $[\mathrm{GL}(2, F)]$-compatible trivialization on $F^2 - 0$. The extensive lower blank remainder and folio 261 are preserved through the authority bitmap and page boundary. The inherited reflow removes this physical topology; it was used only for post-freeze comparison.

Authority physical page 19 / printed 262. Section 4 opens with the Heisenberg-group hypotheses, the three standard assertions (4.1.1)-(4.1.3), and the beginning of Proposition 4.3. The authority uses ω for the central character and ω' for its extension; these symbols and the page break after "à $T'^{\sim}$" are retained. The candidate is broadly parallel in this passage but does not preserve the source page boundary or exact typographic emphasis.

Authority physical page 20 / printed 263. The page completes Proposition 4.3, parametrizes irreducible admissible ε-representations of $T^{\sim}$ by (α_1, α_2), gives the two splittings in 4.4, and states the full global Theorem 4.6. The inherited TeX replaces the authority's global center description and theorem with a different local-model paragraph; that rewriting was rejected. The displayed automorphy condition and all quotient groups follow the authority.

Authority physical page 21 / printed 264. Section 5 begins. The page defines the induced representation and condition (5.2.1), then starts the sheaf-theoretic interpretation in 5.3. The authority's punctuation, the use of B^+ and U^+, and the page-ending hyphen "Finale-" are retained. The candidate reflows the transition across pages and regularizes several source forms; it does not control the diplomatic record.

Authority physical page 22 / printed 265. The page begins with the printed reference “fonctions 2.2.1,” then gives the Jacquet-module exactness statement, the sheaf exact sequence, the adjunction functor r, exact sequence (5.4.1), and integral (5.4.2). The inherited TeX silently changes the printed reference to 5.2.1 and replaces $\rho \circ int_w$ by the looser notation $\rho \cdot w$. The authority’s literal reference and operator notation were preserved.

Authority physical page 23 / printed 266. The page contains (D), 5.5, the regularized intertwining integral (5.5.2), the operator I in (5.5.3), Lemma 5.6, and the first two cases of 5.7. The authority uses the parameters α_1, α_2. In Lemma 5.6(ii) the quoted expression includes the operator I. The candidate changes notation to χ_i and misstates the quoted expression by dropping I and introducing an inverse; the authority controls.

Authority physical page 24 / printed 267. The page completes 5.7(c), states Proposition 5.8 in full, defines the special constituent and principal series in 5.9, and begins the *rational/analytic* treatment of the intertwining operator in 5.10. The inherited TeX truncates Proposition 5.8 and replaces the authority's 5.9-5.10 material by the later Moen theorem, producing a major sequence and content divergence. None of that substitution was accepted. The page ends after the nonzero-scalar assertion.

Authority physical page 25 / printed 268. The page continues 5.10, then visibly repeats the label "5.10." before the Moen result, states Theorem 5.11, gives the complementary-series discussion 5.12, and records Flicker's correspondence in 5.13. The authority has s ∈ R and the normalized operator $I/L(\sigma)$. The inherited TeX changes s to iR and substitutes I for $I/L(\sigma)$; those changes were rejected. The page boundary and all χ_i/α_i notation follow the authority.

Authority physical page 26 / printed 269. The page gives 5.14, Proposition 5.15, formula (5.15.1), 5.16, 5.17, and the opening of 5.18. The authority distinguishes ψ from $\overline{\psi}$: ψ occurs in the equivariance law, while the coinvariant spaces and regularized integral use $\overline{\psi}$. It also prints the functor τ_{ψ}. The inherited TeX drops these bars and corrupts the τ_{ψ} morphism. The source ends with the hyphen “d’entrelace-”.

Authority physical page 27 / printed 270. The page begins with "ment," completing the word broken at the preceding boundary, then gives (5.18.1), the literal cross-reference "2.14," Theorem 5.19, and the $n = 3$ uniqueness sentence. The theorem uses $(W \otimes \overline{\psi})_U$ and the multiplicity $(n-1)/2$. The lower majority of the authority page is intentionally blank; the candidate reflow removes this topology and drops the conjugation bar.

Authority physical page 28 / printed 271. Section 6 opens with the two explicit function models. The authority uses ε^{-1}-functions, the induction subgroup $d_1(F^{*n})d_2(F^*)\mu$, the quotient $d_2(F^*) \cdot U\backslash \mathrm{GL}(2, F)^{\sim}$, and action factors $(\alpha_2 - 1/2)$ and $(\alpha_1 + 1/2)$. The inherited TeX replaces these by an $\varepsilon\chi_1$-model, omits U, and changes both action factors. Those substitutions were rejected after direct formula reinspection.

Authority physical page 29 / printed 272. The page defines the Radon transform and the relation $< \ell, v >= 1$, prints the density $dxdy/d\ell$, and literally reads “menue de Haar.” It then states that the same formula sends functions on $P^\vee$ to functions on P, gives (6.4.1)-(6.4.3), and begins 6.5. The candidate changes $d\ell$ to dt, silently regularizes the printed wording, and misstates the reverse-direction sentence. The authority controls.

Authority physical page 30 / printed 273. The page completes (6.5.1)-(6.5.4), including the multiplicative Haar measures d^*y and $d^*\lambda$, then formulates the V', V'' and R problem in 6.6 and the block decomposition in 6.7. The authority's 6.6 integral is over F^*, and the exceptional character is $||\lambda||^{\pm 1/n}$. The candidate changes the domain and drops $\pm$. The blank lower remainder and the end of section 6 are retained.

Authority physical page 31 / printed 274. The authority opens section 7 and gives the Eisenstein operator, Fourier coefficient, constant term, and Fourier expansion. It prints A and A^{*n} in 7.1, K_∞-finiteness, $\overline{\psi}$ in the Fourier integral, $V_{U,\psi}$, and $\mathrm{diag}(\lambda^2, \lambda)$ in the final sum. The inherited candidate substitutes different domains, drops the conjugation bar, changes the indexed space, and replaces the final matrix. Authority forms control.

Authority physical page 32 / printed 275. The single closing sentence is physically isolated on this page. The authority prints the covered finite-idèle domain $A^{f*\sim}$, the factor $d_1(a^2)d_2(a)$, a semicolon after “décomposable”, and then a substantial blank remainder. The candidate reflows the sentence and changes the domain ; the authority boundary and notation are retained.

Authority physical page 33 / printed 276. This page begins BIBLIOGRAPHIE and contains entries [1]-[12]. Literal source forms are preserved, including “non archimedian”, “Lectures Notes”, “forms- Several”, and “Kodanska”, together with the printed punctuation, line breaks, underlined volume numerals, and folio. The candidate’s bibliographic regularizations are comparison-only.

Authority physical page 34 / printed 277. This terminal page contains entries [13]-[24], followed by the horizontal rule, folio 277, blank remainder, and true physical EOF. Literal source forms include “C.Moen”, “Nancago”, lowercase “grössencharaktere”, and “Springer Lecture Note 189”. The candidate silently normalizes several of these forms and does not preserve the terminal page topology ; the authority controls.

The authority ends with bibliography entries [13]–[24], a horizontal rule, printed folio 277, blank remainder, and physical EOF.